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Improved systematic evaluation of a strontium optical clock with uncertainty below 1×10^{-18}

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Abstract

We report a systematic uncertainty of 9.2×10^{-19} for the Sr1 optical lattice clock at the University of Science and Technology of China (USTC), achieving accuracy at the level required for the roadmap of the redefinition of the SI second. A finite-element model with *in situ*-validated, spatially-resolved chamber emissivity reduced blackbody radiation (BBR) shift uncertainty to 6.3×10^{-19} . Concurrently, the externally mounted lattice cavity, by providing a larger beam waist, reduced the atomic density and thereby suppressed the density shift. Enhanced lattice depth modulation consolidated lattice light shift uncertainty to 6.3×10^{-19} by enabling simultaneous determination of key polarizabilities and magic wavelength. Magnetic shifts were resolved below 10^{-18} via precision characterization of the second-order Zeeman coefficient. Supported by a clock laser stabilized on an ultralow-expansion glass cavity with crystalline-coated mirrors and refined temperature control suppressing BBR fluctuations, the clock also achieves a frequency stability better than 1×10^{-18} at 30 000 s averaging time. These developments collectively establish a new benchmark in USTC Sr1 clock performance and pave the way for high-accuracy applications in metrology and fundamental physics.

1. Introduction

Optical lattice clocks [1–4] and trapped-ion clocks [5–9] have revolutionized precision metrology, emerging as prime candidates for redefining the SI second [10]. These systems employ ultra-stable lasers interrogating ultranarrow atomic transitions, with accuracy fundamentally limited by systematic uncertainties from environmental perturbations. Among multiple prerequisites for SI second redefinition, a key requirement is to deploy at least three optical clocks based on the same reference transition at different institutions, each with a comprehensive uncertainty budget $\leq 2 \times 10^{-18}$. Although this requirement

has not yet been fully met, several systems have recently achieved the uncertainty threshold. For ^{87}Sr clocks, two independent teams have demonstrated uncertainties below 2×10^{-18} [4, 11]. Similarly, two $^{27}\text{Al}^+$ clock teams have also reached this level [5, 12]. Closing this gap necessitates deploying additional clocks with sub- 2×10^{-18} uncertainties.

Based on the 698 nm clock transition, one of the secondary representations of the second, strontium optical lattice clocks currently facilitate the most precise metrology experiments. Their accuracy is constrained by systematic perturbations including external field interactions and interatomic collisions [13, 14], with blackbody radiation (BBR)

and lattice light shifts dominating the uncertainty budget [15–21]. For BBR shifts, while the dynamic term is well-characterized through empirically constrained atomic models [20], uncertainty reduction demands precise knowledge of the radiation field temperature. Although BBR shields improve thermal uniformity [22, 23] and cryogenic environments suppress the radiation field [9, 24–26], accurate in-vacuum temperature determination remains crucial [27]. This is typically achieved through direct probe measurements [14, 28] or validated thermal models [2, 29]. The lattice light shift presents additional complexity, requiring characterization of multiple parameters ranging from empirical correlations [14, 17] to *ab initio* approaches based on fundamental atomic properties [18, 19]. To suppress lattice light shift uncertainty to 1.0×10^{-18} and below, evaluations based on microscopic perspectives—providing interpretations independent of experimental configurations—are essential.

Demonstrating substantial advances beyond our prior evaluation [30], this work achieves a systematic uncertainty of 9.2×10^{-19} for the USTC Sr1 strontium optical clock—meeting one key threshold (2×10^{-18}) required as part of the multi-prerequisite roadmap for SI second redefinition. We address the BBR shift through a comprehensive finite-element (FE) model incorporating experimental parameters for all vacuum chamber surfaces. This framework predicts atomic-position radiation temperatures using surrounding material temperatures and exchange factors, validated against in-vacuum probe data from an isomorphic test chamber and refined by quantifying spatial temperature non-uniformity. Combined with precision temperature control, this approach achieves a BBR shift uncertainty of 6.3×10^{-19} . For lattice light shift evaluation, we implemented an externally mounted cavity lattice providing $60\times$ power enhancement and depths exceeding $800E_r$. The high degree of polarization purity in the linearly polarized lattice minimizes vector light shifts, thereby facilitating precise calibration of the tensor shift component. Through a series of self-comparison measurements, we determined four critical parameters: (i) the electric dipole ($E1$) polarizability derivative, decoupled via a second-order differential protocol by modulating the lattice depth at distinct laser frequencies; (ii) the magic frequency and (iii) the hyperpolarizability, jointly characterized by scanning the lattice laser frequency around the zero-shift point to constrain their values simultaneously; and (iv) the electric quadrupole-magnetic dipole ($E2-M1$) polarizability, evaluated by modulating the axial motional quantum number (n_z) of the atoms. This multi-faceted approach culminated in a consolidated lattice light shift uncertainty of 6.3×10^{-19} . Concurrently, an expanded beam waist suppresses atomic collisions, establishing density shift uncertainty at the 10^{-20} level while significantly reducing

coupling between density effects and lattice shift evaluations. The second-order Zeeman shift for the magnetically insensitive clock transition is also characterized with 10^{-19} -level uncertainty. Integration of these precisely evaluated shifts establishes Sr1's total systematic uncertainty below 1×10^{-18} , satisfying the accuracy requirement for imminent SI-second redefinition.

2. Experimental Setup

The USTC Sr1 optical lattice clock employs a strontium atomic beam generated at 350°C , extensively studied in prior work [31]. Doppler cooling is applied to atoms via the $(5s^2)^1S_0 \leftrightarrow (5s5p)^1P_1$ transition at 461 nm. Sequential deceleration through a Zeeman slower precedes deflection and collimation in a 2D-magneto-optical trap (MOT) before entry into the science chamber (figure 1(a)). Within this chamber, three orthogonal 461 nm laser pairs confine atoms in a blue MOT, achieving mK-level temperatures. Frequency-swept 689 nm beams then drive $(5s^2)^1S_0 \leftrightarrow (5s5p)^3P_1$ transitions: $F' = \frac{11}{2}$ for trapping and $F' = \frac{9}{2}$ for stirring, transferring atoms to a red MOT. This yields μK -level temperatures and enables efficient optical lattice loading at $U_0 = 200E_r$. Key upgrades implemented in this work include: (1) a vacuum-externally mounted optical lattice cavity, (2) enhanced in-lattice cooling protocols, (3) utilization of the magnetically insensitive transition for clock interrogation.

2.1. Externally mounted cavity lattice

Precision interrogation of the ^{87}Sr clock transition requires confining atoms in a one-dimensional optical lattice within the Lamb–Dicke regime, suppressing motion-induced Doppler and photon recoil shifts [32]. As shown in figure 1(a), a vacuum-external optical lattice cavity [17, 33] operating at 813 nm provides $60\times$ intensity enhancement. This configuration expands the lattice beam waist from $37\ \mu\text{m}$ to $155\ \mu\text{m}$ compared to the previous reflective setup, requiring four times less input power. The output mirror (bottom) incorporates a piezo ring chip for active cavity length stabilization. The lattice cavity's input mirror and the clock laser's phase-reference mirror (used for active path-length stabilization) are mounted on a common structure; thereby, their relative phase is stable, preventing the first-order Doppler shift.

A standard Pound–Drever–Hall (PDH) lock maintains lattice cavity resonance by stabilizing the fundamental TEM_{00} mode, where low-frequency feedback drives the cavity piezo and high-frequency components modulate the laser frequency by directly controlling the current of a seed laser, an external-cavity diode laser (ECDL) followed by a fiber amplifier. Another offset-PDH loop operates in parallel, where low-frequency feedback locks the laser PZT to

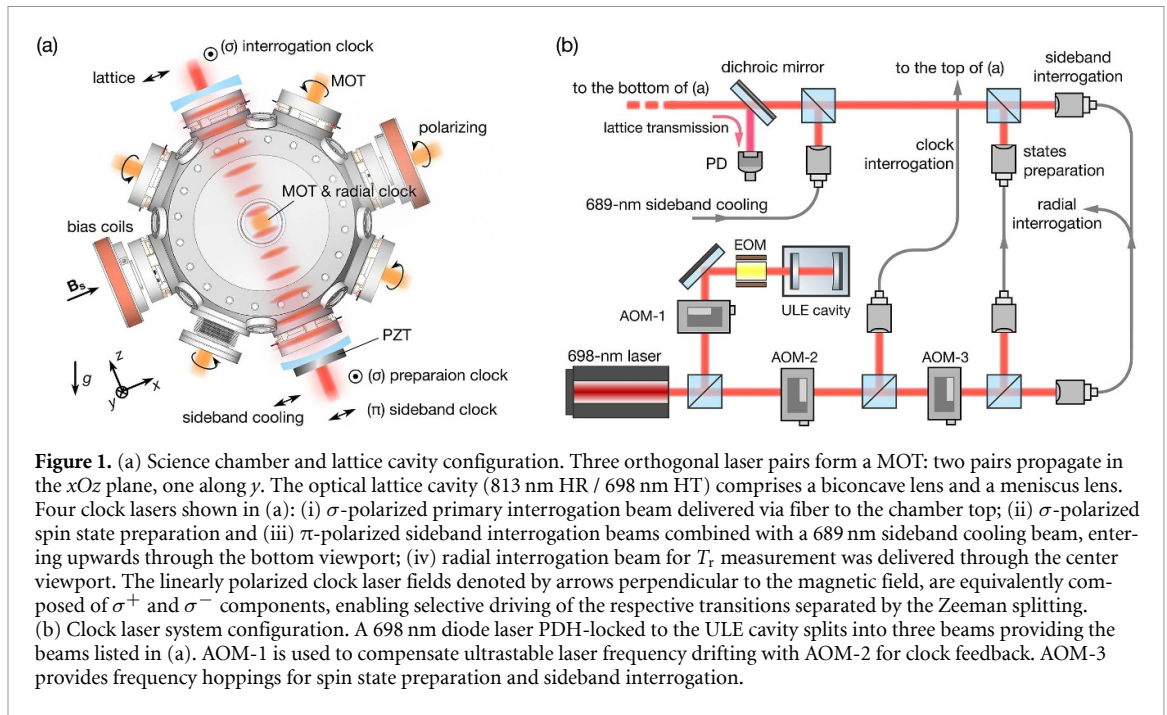


Figure 1. (a) Science chamber and lattice cavity configuration. Three orthogonal laser pairs form a MOT: two pairs propagate in the xOz plane, one along y . The optical lattice cavity (813 nm HR / 698 nm HT) comprises a biconcave lens and a meniscus lens. Four clock lasers shown in (a): (i) σ -polarized primary interrogation beam delivered via fiber to the chamber top; (ii) σ -polarized spin state preparation and (iii) π -polarized sideband interrogation beams combined with a 689 nm sideband cooling beam, entering upwards through the bottom viewport; (iv) radial interrogation beam for T_r measurement was delivered through the center viewport. The linearly polarized clock laser fields denoted by arrows perpendicular to the magnetic field, are equivalently composed of σ^+ and σ^- components, enabling selective driving of the respective transitions separated by the Zeeman splitting. (b) Clock laser system configuration. A 698 nm diode laser PDH-locked to the ULE cavity splits into three beams providing the beams listed in (a). AOM-1 is used to compensate ultrastable laser frequency drifting with AOM-2 for clock feedback. AOM-3 provides frequency hoppings for spin state preparation and sideband interrogation.

a 10 cm ultra-low-expansion (ULE) cavity (finesse 1×10^4), while high-frequency components modulate the offset EOM's bias frequency. This dual-loop architecture suppresses the relative intensity noise (RIN) and achieves a short-term frequency stability of $8 \times 10^{-13}(\tau/s)^{-1/2}$. The long-term drift is maintained below 1 kHz d^{-1} and the absolute frequency is disciplined by an optical frequency comb. These performance characteristics were measured directly at the laser output via an independent beam sampler.

Background lattice light shifts [34] were suppressed using a spectrally purified sum-frequency-generation lattice laser combined with volume Bragg grating spectral filtering, a method established in our previous work on Sr1 with a retro-reflected lattice [35]. This configuration, combined with the narrow transmission bandwidth of the lattice cavity employed in the present work, provides a further suppression of two orders of magnitude, yielding a residual shift safely below 1×10^{-20} .

The linear polarization extinction ratio of the lattice was determined to be approximately 3.7×10^4 , as evaluated by measuring the cavity transmission through a polarizer. The high polarization purity stems directly from the resonance frequency splitting of the two linear polarization eigenmodes, itself caused by birefringence in the intra-cavity optics.

2.2. In-lattice cooling

Three collinear clock laser beams (two $+z$, one $-z$) overlap with the lattice for atomic interrogation (figure 1(a)), enabling spin state preparation, clock operation and sideband spectroscopy. A radially aligned beam (y -direction) scans transverse Doppler profiles. Under a 60 mG bias magnetic field and with

the gradient magnetic field disabled, lattice-trapped atoms undergo simultaneous axial and radial cooling. Axial sideband cooling employs a 689 nm laser spatially overlapped with the lattice to drive the trapping transition [36, 37], achieving $n_z = 0$ population of 0.99 after 50 ms. Concurrent radial cooling via three-axis 689 nm trapping beams [18] operates for 40 ms with optimized detuning and power, minimizing T_r while preserving axial ground-state purity.

Further enhanced cooling is performed by the energy filtering process [38], where the lattice depth undergoes sigmoidal ramping from $200E_r$ to $32E_r$ over 35 ms, holds for 35 ms, then ramps to operational depth. This protocol retains $\sim 10\%$ atomic population after prior cooling stages. The full preparation sequence delivers 1000 atoms within 700 ms for sustained clock operation.

At the lattice depths below $50E_r$ (see figure 2(c)), the measured axial and radial temperature, T_z and T_r , are lower than $0.46 \mu\text{K}$ and $0.30 \mu\text{K}$, respectively. When ramped to depths up to $807E_r$, T_z and T_r scale quasi-linearly to $3.2(1) \mu\text{K}$ and $2.5(2) \mu\text{K}$, with axial excitation attributed to parametric heating from lattice light RIN.

2.3. Magnetically insensitive clock states preparation

To suppress first-order Zeeman shifts, the Sr1 clock employs magnetically insensitive $\sigma^\pm$ -transitions $|g, \pm \frac{5}{2}\rangle \leftrightarrow |e, \pm \frac{3}{2}\rangle$ [39], where e and g represent the excited and ground clock states, respectively. The clock interrogates opposite Rabi lineshape slopes for each of the two symmetric transitions, i.e. requiring four full preparation-interrogation-readout cycles per error signal. These transitions exhibit a factor of 22.4 reduction in first-order Zeeman coefficient

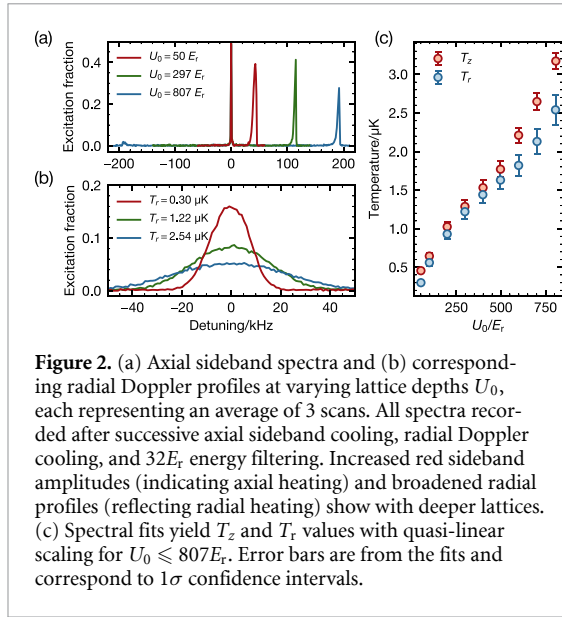


Figure 2. (a) Axial sideband spectra and (b) corresponding radial Doppler profiles at varying lattice depths U_0 , each representing an average of 3 scans. All spectra recorded after successive axial sideband cooling, radial Doppler cooling, and $32E_r$ energy filtering. Increased red sideband amplitudes (indicating axial heating) and broadened radial profiles (reflecting radial heating) show with deeper lattices. (c) Spectral fits yield T_z and T_r values with quasi-linear scaling for $U_0 \leq 807E_r$. Error bars are from the fits and correspond to 1σ confidence intervals.

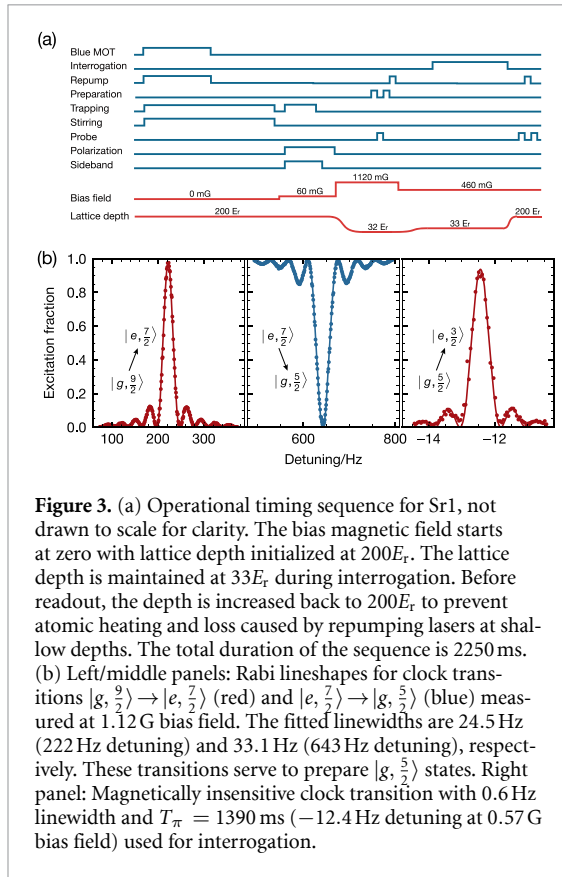


Figure 3. (a) Operational timing sequence for Sr1, not drawn to scale for clarity. The bias magnetic field starts at zero with lattice depth initialized at $200E_r$. The lattice depth is maintained at $33E_r$ during interrogation. Before readout, the depth is increased back to $200E_r$ to prevent atomic heating and loss caused by repumping lasers at shallow depths. The total duration of the sequence is 2250 ms. (b) Left/middle panels: Rabi lineshapes for clock transitions $|g, \frac{9}{2}\rangle \rightarrow |e, \frac{7}{2}\rangle$ (red) and $|e, \frac{7}{2}\rangle \rightarrow |g, \frac{5}{2}\rangle$ (blue) measured at 1.12 G bias field. The fitted linewidths are 24.5 Hz (222 Hz detuning) and 33.1 Hz (643 Hz detuning), respectively. These transitions serve to prepare $|g, \frac{5}{2}\rangle$ states. Right panel: Magnetically insensitive clock transition with 0.6 Hz linewidth and $T_\pi = 1390$ ms (-12.4 Hz detuning at 0.57 G bias field) used for interrogation.

compared to conventional π -transitions between stretched states, significantly suppressing magnetic noise coupling to clock frequency fluctuations.

Figure 1(a) shows the clock laser beams integrated with lattice laser optics at the science chamber, while beam preparation stages including splitting and modulation appear in figure 1(b). State preparation starts simultaneously with sideband and radial cooling (figure 3(a)), where circularly polarized 689 nm light drives stirring transitions to optically pump atoms into $|g, \pm \frac{9}{2}\rangle$ stretched states. During lattice

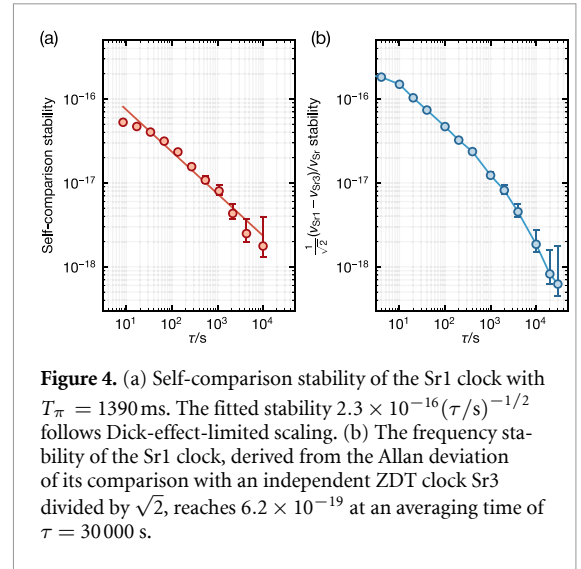


Figure 4. (a) Self-comparison stability of the Sr1 clock with $T_\pi = 1390$ ms. The fitted stability $2.3 \times 10^{-16} (\tau/s)^{-1/2}$ follows Dick-effect-limited scaling. (b) The frequency stability of the Sr1 clock, derived from the Allan deviation of its comparison with an independent ZDT clock Sr3 divided by $\sqrt{2}$, reaches 6.2×10^{-19} at an averaging time of $\tau = 30000$ s.

ramp-down to $32E_r$ for energy filtering, two successive circularly polarized clock pulses transfer atoms from $|g, \pm \frac{9}{2}\rangle$ to $|e, \pm \frac{7}{2}\rangle$ and finally to $|g, \pm \frac{5}{2}\rangle$ states. Magic frequencies for both transitions are independently calibrated, enabling active light shift compensation below 0.1 Hz, negligible compared to spectral widths in figure 3(b). This protocol achieves state purity exceeding 98.5%.

2.4. Clock stability

The Sr1 clock employs Rabi interrogation using a crystalline-coated room-temperature ULE cavity-stabilized laser [40] with π -pulse duration of 1390 ms. During operation, an acousto-optic modulator (AOM) alternately shifts the clock laser frequency to interrogate both $\sigma^\pm$ magnetically insensitive transitions at opposite Rabi lineshape slopes. Excitation fractions are measured via 461 nm fluorescence detection, with a digital servo locking the clock laser frequency to the atomic transition center.

The system achieves fractional frequency stability of $2.3 \times 10^{-16} (\tau/s)^{-1/2}$ in self-comparisons (figure 4(a)), enabling systematic shift evaluation at the low- 10^{-18} level. The operating frequency stability was evaluated by comparison with an independent Sr3 clock. The Allan deviation, plotted in figure 4(b), decreases faster than the typical $(\tau/s)^{-1/2}$ scaling since the Sr3 clock operates in a zero-dead-time (ZDT) mode [41]. At an averaging time of 30 000 s, the Sr1 clock achieves a fractional frequency stability of 6.2×10^{-19} , indicating that the drift of systematic shifts remains below the total uncertainty threshold.

3. Systematic evaluations

Systematic effects perturbing the clock transition must be characterized to determine the unperturbed frequency critical for optical clock accuracy. When couplings between systematic effects and the clock

transition remain stable, active modulation of perturbation sources enables direct measurement of induced frequency shifts. Combining these measurements with precise theoretical models allows determination of both systematic frequency shifts and their associated uncertainties.

This section details our evaluation approach. First, we establish a BBR shift model based on atomic-position equivalent radiation temperature, using dynamic and static polarizabilities to quantify shifts and uncertainties. To extract model parameters to derive operational shifts and uncertainties, we employ active modulation techniques for three dominant perturbations, atomic collisions, lattice light shifts, and Zeeman effects. Finally, we incorporate analyses of minor systematic effects to present the complete uncertainty budget for USTC Sr1 in table 3.

3.1. BBR shift

The BBR shift represents the most significant systematic effect in strontium optical lattice clocks. Its frequency shift is expressed as [20]:

$$\begin{aligned}\Delta\nu_{\text{BBR}} &= -\frac{\Delta\alpha_{\text{stat}}}{2h} \langle E^2 \rangle_{T_a} [1 + \eta(T_a)] \\ &= \Delta\nu_{\text{stat}}(T_a) + \Delta\nu_{\text{dyn}}(T_a),\end{aligned}\quad (1)$$

where $\Delta\alpha_{\text{stat}}$ is the differential static polarizability and $\langle E^2 \rangle_{T_a} = 8.319430(15) (\text{V cm})^{-2} (T_a/300\text{K})^4$ is the mean-square electric field at atomic position temperature T_a , with η denoting the dynamic correction. The shift includes a static component $\Delta\nu_{\text{stat}}$ scaling with T_a^4 and a dynamic component $\Delta\nu_{\text{dyn}}$ with higher-order temperature dependencies.

FE model — we determine T_a using a FE model of the science chamber. Chamber surfaces are partitioned into segments with assigned parameters: emissivity (ϵ), specular reflectivity (R_s), diffuse reflectivity (R_d), and transmissivity (T), satisfying $\epsilon + R_s + R_d + T = 1$. Due to high FE model sensitivity to these parameters, all vacuum components were calibrated at the National Synchrotron Radiation Laboratory to ensure reliability, with measured values incorporated into the model as described in our previous work [42].

Seventeen resistance temperature detectors (RTDs), each with a calibration uncertainty of $\sigma_{\text{RTD}} = 7.3\text{ mK}$, were installed on the science chamber. The steady-state FE solution [43, 44] relates $T_{a,\text{FE}}$ to surrounding chamber surface temperatures T_i :

$$T_{a,\text{FE}}^4 = \sum_{i=1}^{17} F_i T_i^4, \quad (2)$$

where 17 exchange factors $F_i = \partial T_{\text{FE},a} / \partial T_i$ are obtained by perturbing each surface temperature in the FE model with the detailed values already reported in [42]. This configuration enabled the establishment of a FE model (equation (2)) with

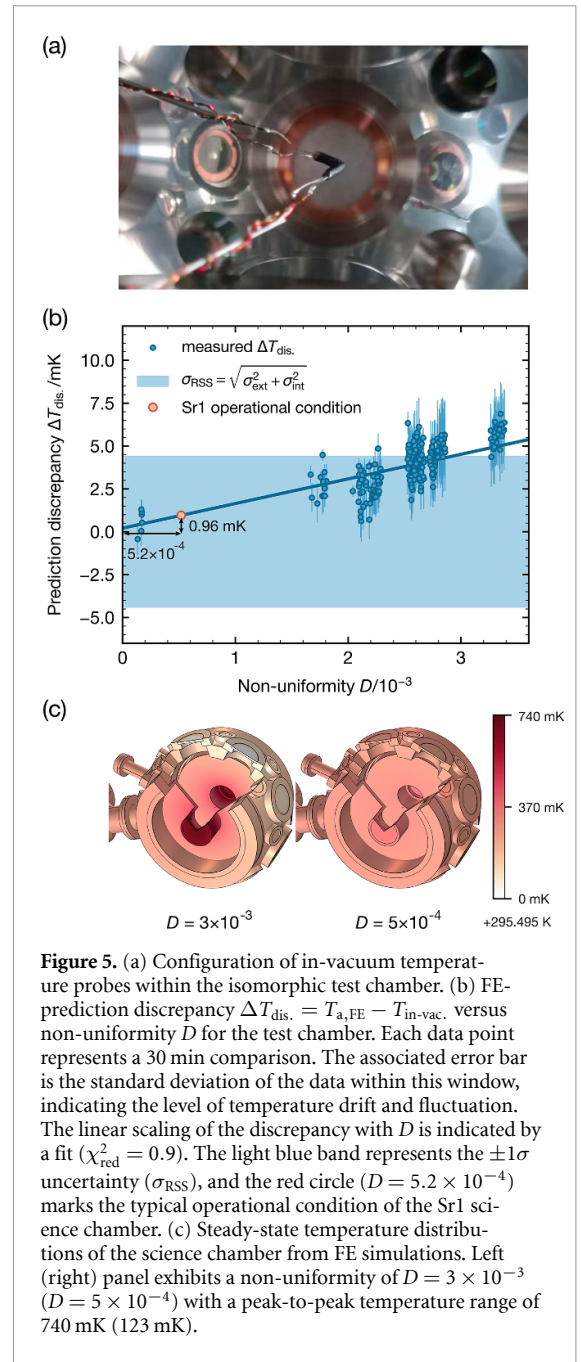


Figure 5. (a) Configuration of in-vacuum temperature probes within the isomorphic test chamber. (b) FE-prediction discrepancy $\Delta T_{\text{dis.}} = T_{a,\text{FE}} - T_{\text{in-vac.}}$ versus non-uniformity D for the test chamber. Each data point represents a 30 min comparison. The associated error bar is the standard deviation of the data within this window, indicating the level of temperature drift and fluctuation. The linear scaling of the discrepancy with D is indicated by a fit ($\chi_{\text{red}}^2 = 0.9$). The light blue band represents the $\pm 1\sigma$ uncertainty (σ_{RSS}), and the red circle ($D = 5.2 \times 10^{-4}$) marks the typical operational condition of the Sr1 science chamber. (c) Steady-state temperature distributions of the science chamber from FE simulations. Left (right) panel exhibits a non-uniformity of $D = 3 \times 10^{-3}$ ($D = 5 \times 10^{-4}$) with a peak-to-peak temperature range of 740 mK (123 mK).

an uncertainty of $\sigma_{\text{ext}} = \sigma_{\text{RTD}} \sqrt{\sum_{i=1}^{17} F_i^2} = 1.9\text{ mK}$ in the predicted $T_{a,\text{FE}}$.

Model validation — we validated the FE model using an isomorphic test chamber equipped with two in-vacuum central RTDs (figure 5(a)), spaced about 5 mm apart, each calibrated against a standard platinum resistance thermometer with an uncertainty of 5.6 mK. The average temperature from these RTDs, $T_{\text{in-vac.}} = (T_{\text{RTD1}} + T_{\text{RTD2}})/2$, was determined with an uncertainty of $\sigma_{\text{int}} = 5.6/\sqrt{2}\text{ mK} = 4.0\text{ mK}$. For the isomorphic test chamber, another seventeen RTDs, with identical geometric configuration to the science chamber and individual calibration uncertainty (7.3 mK, as noted previously), monitored the chamber wall temperatures.

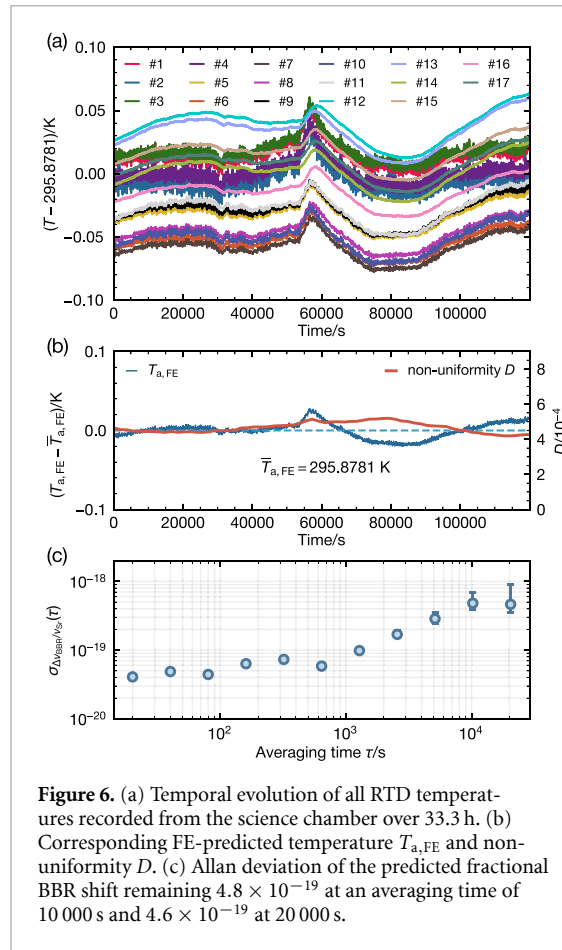


Figure 6. (a) Temporal evolution of all RTD temperatures recorded from the science chamber over 33.3 h. (b) Corresponding FE-predicted temperature $T_{a,FE}$ and non-uniformity D . (c) Allan deviation of the predicted fractional BBR shift remaining 4.8×10^{-19} at an averaging time of 10 000 s and 4.6×10^{-19} at 20 000 s.

For an ideal situation where the chamber wall temperature distribution is fully uniform and constant, T_a equals the chamber temperature. When slight non-uniformity occurs, equation (2) provides a reasonable prediction of T_a . However, prediction deviation increases with significant chamber temperature non-uniformity. By defining the dimensionless non-uniformity D (figure 5(c)) as the weighted standard deviation of all T_i^4 :

$$D^2 = \frac{1}{T_{a,FE}^8} \sum_i F_i (T_i^4 - T_{a,FE}^4)^2, \quad (3)$$

we quantitatively explored this effect using the isomorphic test chamber. The dependence of the prediction discrepancy, $\Delta T_{\text{dis.}} = T_{a,FE} - T_{\text{in-vac.}}$, on non-uniformity D is shown in figure 5(b). Each data point represents a 30 min comparison and exhibits an average temperature drifting rate lower than 9 mK/30 min across all 17 RTDs, thus avoiding prediction ambiguity due to unsteady temperature distribution. The data confirm that the discrepancy $\Delta T_{\text{dis.}}$ scales linearly with non-uniformity D , a relationship characterized by the fit $\Delta T_{\text{dis.}}/\text{K} = 1.4D + 2.1 \times 10^{-4}$ with a reduced chi-square $\chi_{\text{red}}^2 = 0.90$.

For the Sr1 science chamber, owing to the optical table's secondary air conditioning, the average temperature drifting rate of all RTDs remains below 9 mK/30 min. The typical temperature non-uniformity is also below 5.2×10^{-4} (figure 6(b)),

Table 1. Uncertainty contributors to the radiation temperature T_a and uncertainty contributors to the total shift $\Delta\nu_{\text{BBR}}$.

Items	Value
σ_{int}	4.0 mK
σ_{ext}	1.9 mK
T_a	295.8781(44) K
$\sigma_{\text{BBR}, T_a}/\nu_{\text{Sr}}$	3.1×10^{-19}
$\sigma_{\text{BBR}, \text{stat}}/\nu_{\text{Sr}}$	1.3×10^{-19}
$\sigma_{\text{BBR}, \text{dyn}}/\nu_{\text{Sr}}$	5.4×10^{-19}
$\Delta\nu_{\text{BBR}}/\nu_{\text{Sr}}$	$-50\,222.8(6.3) \times 10^{-19}$

and thus the prediction discrepancy is expected to be lower than 1.0 mK, consistent with the root sum square $\sigma_{\text{RSS}} = (\sigma_{\text{int}}^2 + \sigma_{\text{ext}}^2)^{1/2} = 4.4$ mK. Therefore, the FE prediction is validated by the in-vacuum probes, and σ_{RSS} characterizes the final systematic uncertainty of T_a based on the above analysis. The improvement in this result compared with the previous work [42] stems from the installation of two in-vacuum probes and a new quantitative analysis of temperature non-uniformity. The successful agreement between the FE-model prediction and the in-vacuum probe measurements, as demonstrated under typical operating conditions, is a direct outcome of these enhancements.

Clock operation — infrared thermography of peripheral equipment (e.g. CCDs) identified extraneous heat sources, and these were deactivated during clock operation. Optimized thermal management of the MOT coil water-cooling system limited spatial temperature gradients to below 110 mK over 33.3 h of continuous operation (figure 6(a)). The predicted $T_{a,FE}$ exhibited a peak-to-peak variation below 48 mK during this period (figure 6(b)), with the Allan deviation of the resulting fractional BBR shift, $\Delta\nu_{\text{BBR}}/\nu_{\text{Sr}}$, remaining at 4.6×10^{-19} for a 20 000 s averaging time (figure 6(c)). During clock operation, the BBR shift is actively compensated every 1000 s based on the model prediction, further suppressing the degradation of clock stability from temperature fluctuations.

With the determined radiation temperature of $T_a = 295.8781(44)$ K (figure 6(b) and table 1), whose uncertainty of 4.4 mK contributes 0.13 mHz to the frequency uncertainty, the BBR shift is evaluated. The static shift $\Delta\nu_{\text{stat}} = -2.15571(6)$ Hz is obtained from the T_a^4 scaling, with its uncertainty derived from the static polarizability [45], while the dynamic shift is determined to be $\Delta\nu_{\text{dyn}} = -140.16(23)$ mHz with its uncertainty calculated following the direct method in [42]. The total BBR shift of $-50\,222.8 \times 10^{-19}$ is obtained by directly summing the static and dynamic shifts, while its uncertainty of 6.3×10^{-19} is obtained by combining in quadrature the uncertainties from the temperature (σ_{BBR, T_a}), static polarizability ($\sigma_{\text{BBR}, \text{stat}}$), and dynamic component ($\sigma_{\text{BBR}, \text{dyn}}$), as summarized in table 1.

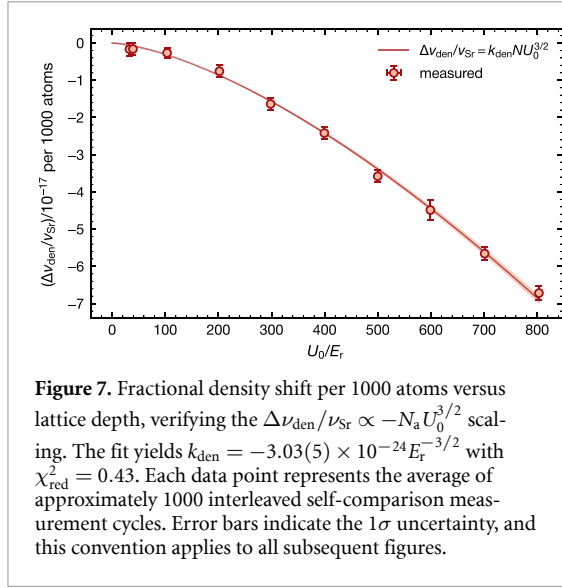


Figure 7. Fractional density shift per 1000 atoms versus lattice depth, verifying the $\Delta\nu_{\text{den}}/\nu_{\text{Sr}} \propto -N_a U_0^{3/2}$ scaling. The fit yields $k_{\text{den}} = -3.03(5) \times 10^{-24} E_r^{-3/2}$ with $\chi_{\text{red}}^2 = 0.43$. Each data point represents the average of approximately 1000 interleaved self-comparison measurement cycles. Error bars indicate the 1σ uncertainty, and this convention applies to all subsequent figures.

3.2. Density Shift

Interatomic collisions within lattice sites induce density shifts perturbing the clock transition frequency and distorting Rabi lineshapes [46, 47]. These shifts originate from residual *s*-wave [48, 49] and *p*-wave scattering [46, 50], modulated by technical factors including clock laser-lattice misalignment [51], excitation fraction [46, 52], and temperature-depth correlations [47]. Experimentally, when the lattice depth is changed, the density shift follows $\Delta\nu_{\text{den}}/\nu_{\text{Sr}} \propto -N_a U_0^{3/2}$ [28, 30, 53], where N_a is the atom number.

To quantify density shifts, we performed self-comparison measurements applying a $\sim 4\times$ lever-arm atom number modulation at different lattice depths. Figure 7 shows the density shift per 1000 atoms versus lattice depth. Across a wide range of lattice depths (33–803 E_r), the measured data are in agreement with the model $\Delta\nu_{\text{den}}/\nu_{\text{Sr}} = k_{\text{den}} N_a U_0^{3/2}$, with a fitted value of $k_{\text{den}} = -3.03(5) \times 10^{-24} E_r^{-3/2}$.

At operational conditions $N_a = 1000(40)$ and $U_0^{\text{op}} = 33.0(1) E_r$, the density shift is $\Delta\nu_{\text{den}}/\nu_{\text{Sr}} = -5.7(2) \times 10^{-19}$. This represents a 30-fold reduction compared to previous work, enabling significant decoupling from lattice light shifts during depth-modulated self-comparisons. We note that while theoretical prediction expects a density shift scaling law of $\Delta\nu_{\text{den}} \propto U_0^{5/4}$ under adiabatic temperature change [47], the measured shifts in our experimental conditions are well described by the scaling law $\propto U_0^{3/2}$, as evidenced by a reduced chi-square of $\chi_{\text{red}}^2 = 0.43$. One possible explanation for this deviation, proposed in [19], involves higher-order contributions arising from atomic interactions that are mediated by off-resonant lattice scattering.

3.3. Lattice light shift

Optical lattices confine strontium atoms at standing-wave anti-nodes by minimizing AC Stark shifts. Strontium clocks must operate at the magic frequency

where $E1$ polarizabilities of clock states balance, yielding near-zero net shift [54, 55]. For experimental implementation, we first align the bias magnetic field parallel to lattice polarization to suppresses vector light shifts and minimizes tensor shift sensitivity to magnetic field angle fluctuations. Following recent theoretical advances, we apply the comprehensive higher-order model for lattice light shifts [18, 21, 56]:

$$\begin{aligned} \Delta\nu_l = & \left[\frac{\partial \alpha_{E1}}{\partial \nu} (\nu - \nu_{E1}^m) - \alpha_{\text{qm}} \right] \left(n_z + \frac{1}{2} \right) \bar{u}^{1/2} \\ & - \left[\frac{\partial \alpha_{E1}}{\partial \nu} (\nu - \nu_{E1}^m) + \frac{3}{2} \beta \left(n_z^2 + n_z + \frac{1}{2} \right) \right] \bar{u} \\ & + 2\beta \left(n_z + \frac{1}{2} \right) \bar{u}^{3/2} - \beta \bar{u}^2, \end{aligned} \quad (4)$$

where $\bar{u} = u[1 + jk_B T_r / (u E_r)]^{-1}$ accounts for radial motion averaging at temperature T_r . Here $u = U_0/E_r$ is dimensionless depth, n_z the axial quantum number, ν the lattice frequency, and ν_{E1}^m the $E1$ magic frequency. Coefficients include α_{E1} ($E1$ polarizability), α_{qm} ($E2$ – $M1$ polarizability), and β (hyperpolarizability), all in frequency units.

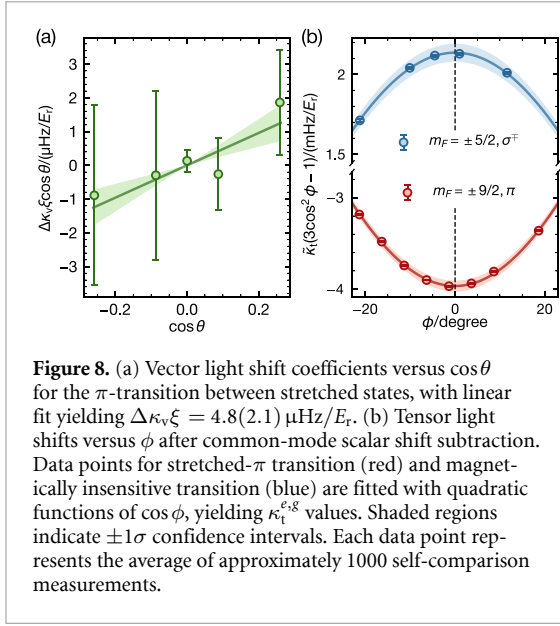
Calibration of $E1$ vector and tensor light shifts — prior to determining the coefficients in equation (4), the $E1$ vector and tensor light shifts must be calibrated. For the $m_F = m_{e(g)}$ state, these shifts are given by [57]:

$$\begin{aligned} \Delta\nu_{v+t}^{e(g)} = & m_{e(g)} \kappa_v^{e(g)} \xi \cos \theta U_0 \\ & + \kappa_t^{e(g)} \left[3m_{e(g)}^2 - F(F+1) \right] \\ & \times (3|\vec{\epsilon}_1 \cdot \hat{\mathbf{B}}_s|^2 - 1) U_0, \end{aligned} \quad (5)$$

where κ_v and κ_t are the vector and tensor polarizabilities, θ is the angle between the lattice wave vector and the bias magnetic field direction $\hat{\mathbf{B}}_s$, and $\vec{\epsilon}_1$ is the complex polarization vector with ξ as the polarization parameter. For ideal linear polarization, $\xi = 0$ and $(3|\vec{\epsilon}_1 \cdot \hat{\mathbf{B}}_s|^2 - 1) = (3 \cos^2 \phi - 1)$, with ϕ being the angle between $\vec{\epsilon}_1$ and $\hat{\mathbf{B}}_s$. As detailed in section 2, the Sr1 apparatus adopts a conventional configuration: the lattice wavevector is perpendicular to the bias magnetic field ($\theta = \pi/2$), and the linearly lattice polarization aligns parallel to the magnetic field ($\phi = 0$).

We first calibrated the $E1$ vector light shifts. By varying the bias magnetic field orientation within the $\mathbf{k}_l$ – $\vec{\epsilon}_1$ plane (where $\mathbf{k}_l$ denotes the lattice wave vector) and applying a lattice depth modulation of $\Delta U_0 = 450 E_r$, we measured these shifts through symmetric sublevel transition splitting, as shown in figure 8(a). This yielded a minimal vector shift coefficient $\Delta\kappa_v \xi = 4.8(2.1) \mu\text{Hz}/E_r$, confirming pure linear lattice polarization with $|\Delta\kappa_v| = 0.22(5) \text{Hz}/E_r$ [57].

Angular-dependent tensor light shifts were measured through the average frequency of symmetric sublevel transitions. From fits to the tensor shifts



of the stretched- π and the magnetically insensitive transition (figure 8(b)), we extracted $\kappa_t^g = -6.4(2.7) \mu\text{Hz}/E_r$ and $\kappa_t^e = -61.5(2.4) \mu\text{Hz}/E_r$. By determining the symmetry axis of the angular dependence of the tensor light shifts (figure 8(b)), we ensured that the magnetic field orientation was parallel to the lattice polarization, with an alignment uncertainty of 0.12° which accounts for both the fitting uncertainty in figure 8(b) and the residual misalignment perpendicular to the scanning plane.

Parameter determination — characterization of lattice light shift requires measurement of four parameters in equation (4): $\partial \alpha_{E1}/\partial \nu$, ν_{E1}^m , α_{qm} , and β . During lattice depth modulation, frequency shifts arising from atomic density variations are corrected using the calibrated model to isolate the lattice light shift. The associated uncertainty of this correction is included in the total uncertainty as an independent component for each data point.

For $\partial \alpha_{E1}/\partial \nu$, we employed differential self-comparison at two lattice frequencies (ν_1, ν_2):

$$\delta[\Delta \nu_l]_{\nu_2} - \delta[\Delta \nu_l]_{\nu_1} = -\frac{\partial \alpha_{E1}}{\partial \nu} (\nu_2 - \nu_1) \delta[u_z], \quad (6)$$

where $\delta[\cdot]$ denotes differences between modulated ($u_z^{\text{mod}} = 49\text{--}572$) and reference ($u_z^{\text{ref}} = 89$) conditions, with $u_z = \bar{u} - (n_z + 1/2)u^{1/2}$. At about ± 1 GHz detuning around ν_{E1}^m , we obtained $\partial \alpha_{E1}/\partial \nu = 1.811(5) \times 10^{-11}$ (figure 9(a)) with a reduced chi-square $\chi_{\text{red}}^2 = 1.22$ (throughout this article, all uncertainties were scaled by $(\chi_{\text{red}}^2)^{1/2}$ when $\chi_{\text{red}}^2 > 1$). We resolved β and ν_{E1}^m by modulating $\bar{u}$ from 32 to 792 at four frequencies within 150 MHz of ν_{E1}^m (figure 9(b)).

Determination of α_{qm} required n_z state modulation during self-comparisons. The $n_z = 1$ state was prepared through: blue sideband excitation $|g, n_z = 0\rangle \leftrightarrow |e, n_z = 1\rangle$ at $60E_r$; removal of residual $|g, n_z = 0\rangle$ atoms via 461 nm light; carrier π -pulse transfer $|e, n_z = 1\rangle \rightarrow |g, n_z = 1\rangle$; and finally this achieved $\langle n_z \rangle = 1.02(5)$ with maintained spin purity. Self-comparison results between $n_z = 0$ (reference) and $n_z = 1$ states appear in figure 9(c). Then iterative fitting determined β , ν_{E1}^m , and α_{qm} values as follows.

We performed global iterative fitting [18] of all modulation data [depth and frequency in figure 9(b), n_z state in (c)] using equation (4), propagating uncertainties in $\partial \nu \alpha_{E1}$, U_0 , T_r , ν , and $\langle n_z \rangle$. Converged results yield: $\beta = -0.466(13) \mu\text{Hz}$, $\nu_{E1}^m = 368\,554\,826.6(5) \text{ MHz}$, $\alpha_{qm} = -1.057(32) \text{ mHz}$. Table 2 compares these values with literature, noting the ~ 362 MHz discrepancy in ν_{E1}^m relative to [18] originates from different magnetic sublevel transitions. The determined α_{qm} value is further supported by consistency with [58] and theoretical predictions from [59].

Operational shift evaluation — at operational conditions $\nu = \nu_{E1}^m$, $\langle n_z \rangle = 0.00(1)$, $U_0^{\text{op}} = 33.0(1)E_r$ yielding $\bar{u}^{\text{op}} = 32.4(1)$, the lattice light shift is $\Delta \nu_l/\nu_{Sr} = 80.5(63) \times 10^{-19}$. The 6.3×10^{-19} uncertainty is dominated by the 0.47 MHz uncertainty in ν_{E1}^m .

3.4. Zeeman Shift

Residual first-order effects from magnetic field fluctuations synchronized with clock cycles are bounded by:

$$\Delta \nu_{Z1, \text{res}} \leq \langle S_{2n} - S_{2n-1} \rangle / 8, \quad (7)$$

where S_n denotes Zeeman splitting between symmetric transitions during the n th cycle. Using symmetric clock transitions between stretched states, this shift for Sr1 is $0 \pm 8 \times 10^{-19}$ [30]. Magnetically insensitive clock transitions offer a suppression factor of about $22.4 \times$, and thus the uncertainty of $\Delta \nu_{Z1, \text{res}}$ achieves 0.4×10^{-19} .

The second-order Zeeman shift scales quadratically with magnetic field strength and remains unidirectional for all magnetic sublevels [14, 57]:

$$\Delta \nu_{Z2} = \xi_i \Delta \nu_{Z1}^2, \quad (8)$$

where ξ_i is the second-order coefficient for magnetically insensitive transitions. Here $\Delta \nu_{Z1}$ is derived from measured symmetric transition splittings $\Delta \nu_{\text{meas}}$ as $\Delta \nu_{Z1} = \Delta \nu_{\text{meas}} - \Delta \nu_v$, with $\Delta \nu_v$ representing vector light shifts.

We characterized ξ_i through self-comparison measurements with a 500 mG reference bias field and 700–1500 mG modulation fields. Vector light shift was independently quantified as $\Delta \nu_v/U_0 =$

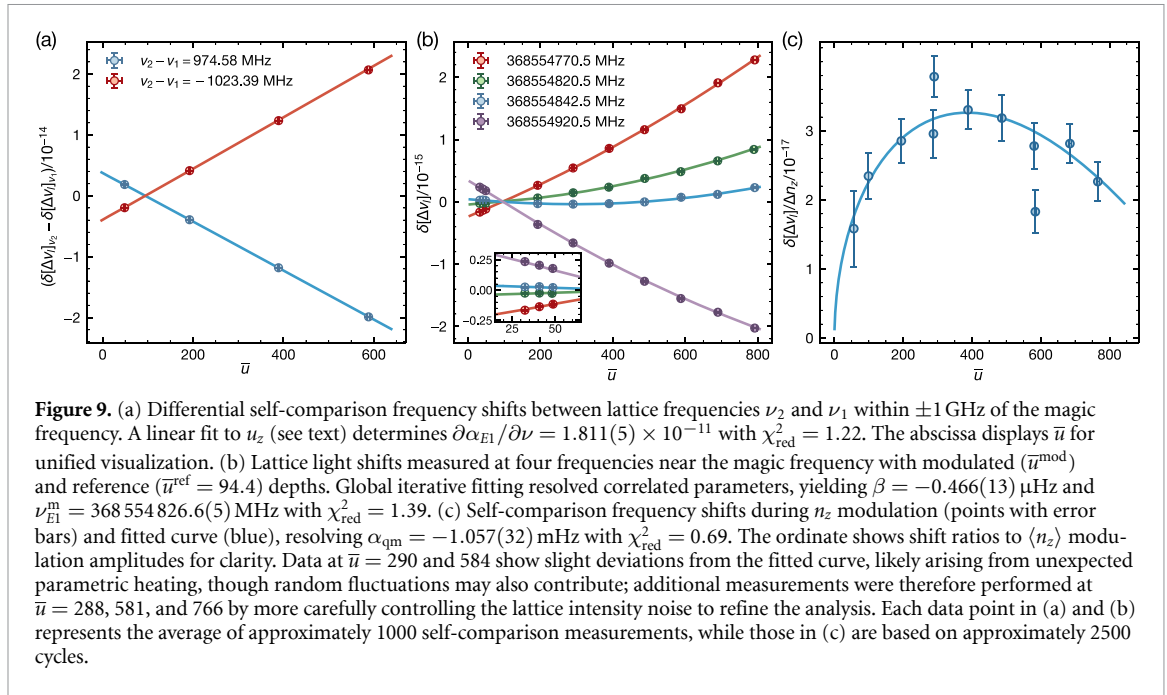


Figure 9. (a) Differential self-comparison frequency shifts between lattice frequencies ν_2 and ν_1 within ± 1 GHz of the magic frequency. A linear fit to u_z (see text) determines $\partial\alpha_{E1}/\partial\nu = 1.811(5) \times 10^{-11}$ with $\chi^2_{\text{red}} = 1.22$. The abscissa displays $\bar{u}$ for unified visualization. (b) Lattice light shifts measured at four frequencies near the magic frequency with modulated ($\bar{u}^{\text{mod}}$) and reference ($\bar{u}^{\text{ref}} = 94.4$) depths. Global iterative fitting resolved correlated parameters, yielding $\beta = -0.466(13)$ μHz and $\nu_{E1}^m = 368\,554\,826.6(5)$ MHz with $\chi^2_{\text{red}} = 1.39$. (c) Self-comparison frequency shifts during n_z modulation (points with error bars) and fitted curve (blue), resolving $\alpha_{\text{qm}} = -1.057(32)$ mHz with $\chi^2_{\text{red}} = 0.69$. The ordinate shows shift ratios to $\langle n_z \rangle$ modulation amplitudes for clarity. Data at $\bar{u} = 290$ and 584 show slight deviations from the fitted curve, likely arising from unexpected parametric heating, though random fluctuations may also contribute; additional measurements were therefore performed at $\bar{u} = 288, 581$, and 766 by more carefully controlling the lattice intensity noise to refine the analysis. Each data point in (a) and (b) represents the average of approximately 1000 self-comparison measurements, while those in (c) are based on approximately 2500 cycles.

Table 2. Parameters of lattice light shifts: $E1$ polarizability derivative ($\partial\nu\alpha_{E1}$), $E1$ magic frequency (ν_{E1}^m), $E2$ – $M1$ polarizability (α_{qm}), and hyperpolarizability (β), compared with literature values.

Quantities	This work	[21]	[18]
$\partial\nu\alpha_{E1}/10^{-11}$	1.811(5)	1.859(5)	1.735(13)
ν_{E1}^m/MHz	368 554 826.6(5)	368 554 825.9(4)	368 554 465.1(10)*
$\alpha_{\text{qm}}/\text{mHz}$	−1.057(32)	−1.24(5)	−0.962(40)
$\beta/\mu\text{Hz}$	−0.466(13)	−0.51(4)	−0.461(14)

* The ν_{E1}^m value from [18] corresponds to π -transition of stretched states and therefore deviates from the other two.

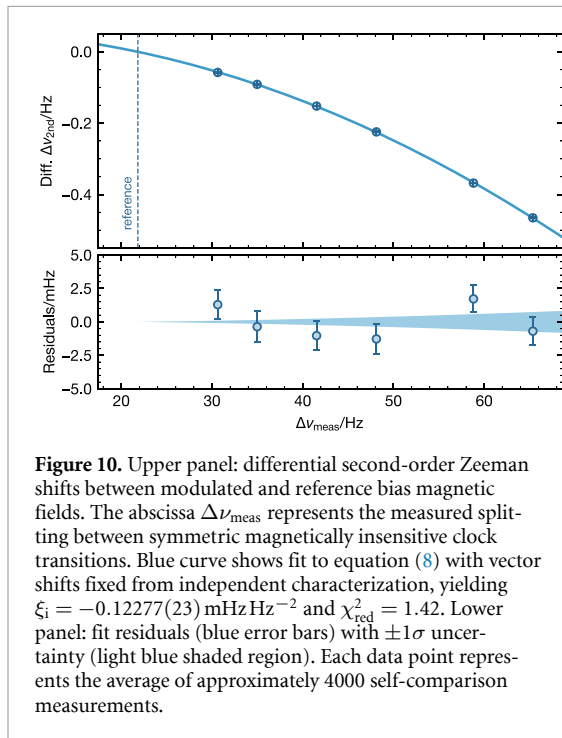


Figure 10. Upper panel: differential second-order Zeeman shifts between modulated and reference bias magnetic fields. The abscissa $\Delta\nu_{\text{meas}}$ represents the measured splitting between symmetric magnetically insensitive clock transitions. Blue curve shows fit to equation (8) with vector shifts fixed from independent characterization, yielding $\xi_i = -0.12277(23)$ mHz Hz $^{-2}$ and $\chi^2_{\text{red}} = 1.42$. Lower panel: fit residuals (blue error bars) with $\pm 1\sigma$ uncertainty (light blue shaded region). Each data point represents the average of approximately 4000 self-comparison measurements.

−7.6(2.8) $\mu\text{Hz}/E_r$ for magnetically insensitive transitions. Constraining $\Delta\nu_v$ in fits to equation (8), we obtained $\xi_i = -0.12277(23)$ mHz Hz $^{-2}$ (figure 10).

At operational conditions ($\Delta\nu_{Z1} = 20.0$ Hz, $B_s^{\text{op}} \approx 460$ mG), the second-order Zeeman shift is $\Delta\nu_{Z2}/\nu_{\text{Sr}} = -1144.1(21) \times 10^{-19}$.

3.5. Other minor systematic shifts

Residual DC Stark shift — this shift originates from static electric fields due to charge accumulation on viewports [60]. As reported in our previous work [30], the y -component of the field (figure 1(a)) caused a shift of $1.4(5.2) \times 10^{-21}$. Given $\Delta\nu_{\text{DC}} \propto E_{\text{stat}}^2$ scaling and $E_{\text{stat}} \propto r^{-2}$ dependence, the shorter y -axis viewport distance (142 mm vs 237 mm for x, z) would yield an $8 \times$ larger shift along y . Electrostatic shielding further suppresses x, z -axis fields by a factor of 3 according to FE simulations. Combining these, we constrain the total shift to $0.0(0.1) \times 10^{-19}$.

Background gas collisions — two-body collisions with residual H_2 induce shifts proportional to loss rate. Using the coefficient $-3.0(3) \times 10^{-17}$ Hz/(atom s $^{-1}$) [61] and measured vacuum lifetime 45(2) s, we obtain a shift of $-6.7(0.7) \times 10^{-19}$.

AOM phase chirp — thermal transients during AOM activation introduce phase chirp, effectively generating Doppler shifts. In Sr1, the AOM resides within the

Table 3. USTC Sr1 uncertainty budget.

Systematic	Shift / 10^{-19}	Uncertainty / 10^{-19}
BBR	−50 222.8	6.3
Density	−5.7	0.2
Lattice	80.5	6.3
1 st Zeeman	0	0.4
2nd Zeeman	−1144.1	2.1
DC Stark	0	<0.1
Background gas	−6.7	0.7
AOM phase chirp	0	<0.1
Line pulling	0	<0.1
Lattice tunneling	0	<0.1
Probe AC Stark	−0.3	<0.1
Servo error	0	0.7
Total	−51,299.1	9.2

fiber noise cancellation loop, with feedback stabilizing the optical path within 20 μ s. Using IQ demodulation to extract the instantaneous phase, we calculated the corresponding frequency shift below 1×10^{-20} .

Line-pulling effect — residual ground-state atoms in non-interrogated sublevels perturb Rabi lineshapes. For Sr1's π -pulse duration ($T_i^{\text{op}} = 1390$ ms) and 98.5% spin purity, uniform residual population leads to a shift $< 1 \times 10^{-21}$, with the worst case $\leq 9 \times 10^{-21}$. Thus we assign a systematic shift of $0.0(0.1) \times 10^{-19}$.

Lattice tunneling — the lattice geometry, tilted at 22.5° relative to gravity (figure 1(a)), yields a tunneling-induced frequency shift (62) below 1×10^{-20} for an interrogation pulse duration $T_i^{\text{op}} = 1390$ ms at a depth of $U_0^{\text{op}} = 33.0(1)E_r$.

Probe AC Stark shift — the clock laser induces a weak ac Stark shift. Based on the coefficient from [63] and the operational parameters, this shift is calculated to be $-2.8(2) \times 10^{-20}$, which accounts for the combined intensity of both polarization components [4].

Servo error — this shift originates from loop delay $T_s = -0.29(2)$ s and clock laser frequency drift [30]. Automatic recalibration and compensation every 1000 s suppress the drift rate to $0.0(0.1)$ mHzs $^{-1}$, bounding the servo error to $0.0(0.7) \times 10^{-19}$.

4. Summary

This work reports systematic upgrades to the USTC Sr1 optical lattice clock, achieving a total systematic uncertainty of 9.2×10^{-19} . A crystalline-coated ULE cavity provides clock stability of 6.2×10^{-19} at 30 000 s averaging time. Refined BRR shift modeling, validated by in-vacuum probes and non-uniformity analysis, constrains its uncertainty contribution to 6.3×10^{-19} . Atomic density shifts are suppressed to -5.7×10^{-19} with 0.2×10^{-19} uncertainty through an expanded-beam-waist optical lattice implementation. Comprehensive lattice light shift calibration

reaches 6.3×10^{-19} uncertainty via strategic self-comparison measurements that determine magic frequency, $E1$ polarizability, hyperpolarizability, and $E2$ – $M1$ polarizability. These advancements integrate with a characterized second-order Zeeman coefficient for magnetically insensitive transitions to collectively achieve the reported uncertainty level.

This performance meets the 2×10^{-18} single-clock accuracy requirement for redefining the SI second, with potential applications in relativistic geodesy and high-resolution dark matter searches. Future optical clocks targeting uncertainties in the low- 10^{-19} range may benefit from combining cryogenic environments [25] with vertical shallow lattice geometries [21], which promise further suppression of BBR and lattice light shift uncertainties.

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References

- [1] Ludlow A D, Boyd M M, Ye J, Peik E and Schmidt P O 2015 *Rev. Mod. Phys.* **87** 637–701
- [2] McGrew W F et al 2018 *Nature* **564** 87
- [3] Takamoto M, Ushijima I, Ohmae N, Yahagi T, Kokado K, Shinkai H and Katori H 2020 *Nat. Photon.* **14** 411
- [4] Aeppli A, Kim K, Warfield W, Safronova M S and Ye J 2024 *Phys. Rev. Lett.* **133** 023401
- [5] Marshall M C et al 2025 *Phys. Rev. Lett.* **135** 033201
- [6] Zhang Z, Arnold K J, Kaewuam R and Barrett M D 2023 *Sci. Adv.* **9** eadg1971
- [7] Tofful A et al 2024 *Metrologia* **61** 045001
- [8] Haussner H N et al 2025 *Phys. Rev. Lett.* **134** 023201
- [9] Zhang B et al 2026 Liquid-nitrogen-cooled $^{40}\text{Ca}^{+}$ ion optical clock with a systematic uncertainty of 4.4×10^{-19} *Phys. Rev. Lett.* **136** 053202
- [10] Dimarcq N et al 2024 *Metrologia* **61** 012001
- [11] Lu X-T, Guo F, Liu Y-Y, Cao J, Li J-A, Xia J-J, Xu Q-F, Lu B-Q, Wang Y-B and Chang H 2025 *Metrologia* **62** 035007
- [12] Ma Z Y et al 2024 *Phys. Rev. Appl.* **21** 044017
- [13] Ludlow A D, Boyd M M, Zelevinsky T, Foreman S M, Blatt S, Notcutt M, Ido T and Ye J 2006 *Phys. Rev. Lett.* **96** 033003
- [14] Bothwell T, Kedar D, Oelker E, Robinson J M, Bromley S L, Tew W L, Ye J and Kennedy C J 2019 *Metrologia* **56** 065004
- [15] Westergaard P G, Lodewyck J, Lorini L, Lecallier A, Burt E A, Zawada M, Millo J and Lemonde P 2011 *Phys. Rev. Lett.* **106** 210801
- [16] Safronova M S, Porsev S G, Safronova U I, Kozlov M G and Clark C W 2013 *Phys. Rev. A* **87** 012509
- [17] Brown R C et al 2017 *Phys. Rev. Lett.* **119** 253001
- [18] Ushijima I, Takamoto M and Katori H 2018 *Phys. Rev. Lett.* **121** 263202
- [19] Nemitz N, Jørgensen A A, Yanagimoto R, Bregolin F and Katori H 2019 *Phys. Rev. A* **99** 033424
- [20] Lisdat C, Drscher S, Nosske I and Sterr U 2021 *Phys. Rev. Res.* **3** L042036
- [21] Kim K, Aeppli A, Bothwell T and Ye J 2023 *Phys. Rev. Lett.* **130** 113203
- [22] Zhang P, Cao J, Yuan J-B, Liu D-X, Yuan Y, Wei Y-F, Shu H-L and Huang X-R 2021 *Metrologia* **58** 035001
- [23] Zhang A et al 2022 *Metrologia* **59** 065009
- [24] Nosske I, Vishwakarma C, Lücke T, Rahm J, Poudel N, Weyers S, Benkler E, Dörscher S and Lisdat C 2025 *Quantum Sci. Technol.* **10** 045076
- [25] Hassan Y S et al 2025 *Phys. Rev. Lett.* **135** 063402
- [26] Ushijima I, Takamoto M, Das M, Ohkubo T and Katori H 2015 *Nat. Photon.* **9** 185
- [27] Jin T, Zhang T, Luo L, Liu L, Zhou M and Xu X 2023 *Measurement* **216** 112946
- [28] Nicholson T et al 2015 *Nat. Commun.* **6** 6896
- [29] Lu B-K, Sun Z, Yang T, Lin Y-G, Wang Q, Li Y, Meng F, Lin B-K, Li T-C and Fang Z-J 2022 *Chin. Phys. Lett.* **39** 080601
- [30] Li J et al 2024 *Metrologia* **61** 015006
- [31] Li J et al 2023 *Rev. Sci. Instrum.* **94** 093202
- [32] Katori H 2011 *Nat. Photon.* **5** 203
- [33] Wang C, Yao Y, Shi H, Yu H, Ma L and Jiang Y 2024 *Chin. Phys. B* **33** 030601
- [34] Fasano R, Chen Y, McGrew W, Brand W, Fox R and Ludlow A 2021 *Phys. Rev. Appl.* **15** 044016
- [35] Jia Z-P, Cui X-Y, Xie Y-J, Zhang X, Niu G-Z, Liu X-Y, Zhu Q-Q, Li J and Dai H-N 2025 *Phys. Rev. Appl.* **23** 014014
- [36] Nemitz N, Ohkubo T, Takamoto M, Ushijima I, Das M, Ohmae N and Katori H 2016 *Nat. Photon.* **10** 258–61
- [37] Bishof M N 2014 Understanding atomic interactions in an optical lattice clock and using them to study many-body physics *PhD Thesis* University of Colorado
- [38] Falke S et al 2014 *New J. Phys.* **16** 073023
- [39] Oelker E et al 2019 *Nat. Photon.* **13** 714
- [40] Zhu X-Q et al 2024 *Rev. Sci. Instrum.* **95** 083002
- [41] Liu X-Y et al 2025 *Phys. Rev. Lett.* **135** 263402
- [42] Yu H, Liu P, Li Y, Jia Z, Zhang X, Yan J, Li J, Dai H and Chen Y 2025 *Measurement* **257** 118527
- [43] Doležal M et al 2015 *Metrologia* **52** 842
- [44] Xiong D, Zhu Q, Wang J, Zhang A, Tian C, Wang B, He L, Xiong Z and Lyu B 2021 *Metrologia* **58** 035005
- [45] Middelmann T, Falke S, Lisdat C and Sterr U 2012 *Phys. Rev. Lett.* **109** 263004
- [46] Bishof M, Martin M J, Swallows M D, Benko C, Lin Y, Quémener G, Rey A M and Ye J 2011 *Phys. Rev. A* **84** 052716
- [47] Swallows M D, Martin M J, Bishof M, Benko C, Lin Y, Blatt S, Rey A M and Ye J 2012 *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **59** 416–25
- [48] Campbell G K et al 2009 *Science* **324** 360–3
- [49] Swallows M D, Bishof M, Lin Y, Blatt S, Martin M J, Rey A M and Ye J 2011 *Science* **331** 1043–6
- [50] Lemke N D, von Stecher J, Sherman J A, Rey A M, Oates C W and Ludlow A D 2011 *Phys. Rev. Lett.* **107** 103902
- [51] Zhou Y-H, Zhang X-F and Wang T 2023 *Phys. Rev. A* **108** 033304
- [52] Ludlow A D, Lemke N D, Sherman J A, Oates C W, Quémener G, von Stecher J and Rey A M 2011 *Phys. Rev. A* **84** 052724
- [53] Lu X, Guo F, Wang Y, Xu Q, Zhou C, Xia J, Wu W and Chang H 2023 *Metrologia* **60** 015008
- [54] Katori H, Takamoto M, Pal'chikov V G and Ovsiannikov V D 2003 *Phys. Rev. Lett.* **91** 173005
- [55] Takamoto M, Hong F-L, Higashi R and Katori H 2005 *Nature* **435** 321
- [56] Bothwell T et al 2025 *Phys. Rev. Lett.* **134** 033201
- [57] Shi C, Robyr J-L, Eismann U, Zawada M, Lorini L, Le Targat R and Lodewyck J 2015 *Phys. Rev. A* **92** 012516
- [58] Dörscher S, Klose J, Maratha Palli S and Lisdat C 2023 *Phys. Rev. Res.* **5** 1012013
- [59] Wu F-F, Shi T-Y, Ni W-T and Tang L-Y 2023 *Phys. Rev. A* **108** 1051101
- [60] Lodewyck J, Zawada M, Lorini L, Gurov M and Lemonde P 2012 *IEEE Trans. Ultrason. Ferroelectr. Freq. Control* **59** 411–5
- [61] Alves B X R, Foucault Y, Vallet G and Lodewyck J 2019 *2019 Joint Conf. IEEE Int. Frequency Control Symp. and European Frequency and Time Forum (EFTF/IFC)* (IEEE) pp 1–2 (<https://doi.org/10.1109/fcs.2019.8856042>)
- [62] Lemonde P and Wolf P 2005 *Phys. Rev. A* **72** 033409
- [63] Xu Q, Lu X, Xia J, Wang Y and Chang H 2021 *Appl. Phys. Lett.* **119** 101105